

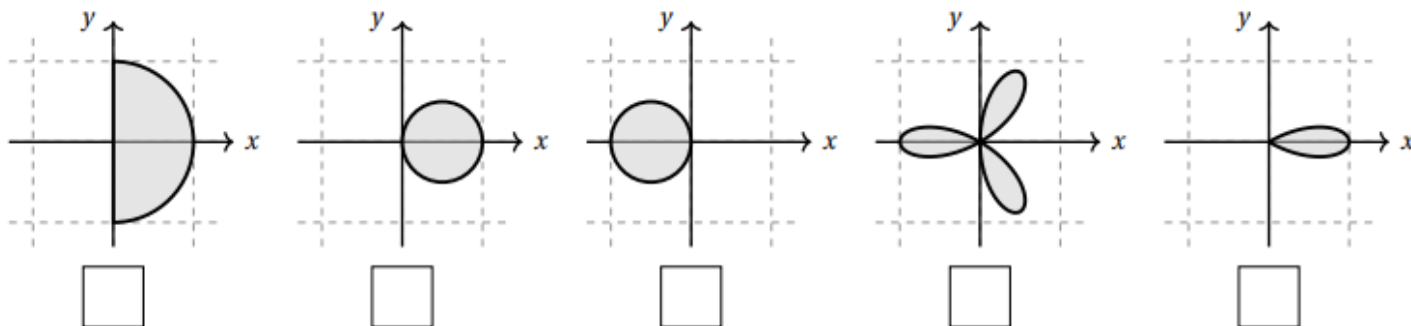
1. Integrate $f(x, y) = x^2 + y^2$ over the domain D , where D is the region bounded by $y = 0$, $x = 1$, & $y = x$.

2. Find the volume of the solid that is bounded above by the cylindrical surface $z = x^2$ and below by the domain enclosed by the parabola $y = 2 - x^2$ and the line $y = x$.

3. Use a double integral to determine the area of the region given by $x^2 + y^2 \geq 1$ & $x^2 + y^2 \leq 4$ which exists above the x -axis.
4. A cylindrical drill bit is used to bore a vertical hole through the center of the solid bounded above by the graph of $f(x, y) = 25e^{-(x^2+y^2)/4}$ and bounded below by the region in the xy -plane given by $x^2 + y^2 \leq 16$. What is the radius of the drill bit if 10% of the volume of the solid is removed?

5. The integral $\iint_R (2x^2 + 2y^2 + y) dA$ has the form $\int_{-\pi/2}^{\pi/2} \int_0^{\cos(\theta)} ?? dr d\theta$ when converted into polar coordinates.

a. Mark the box below with the picture of the region that represents R .



b. Fill in the missing integrand to convert this integral into polar coordinates.

$$\iint_R (2x^2 + 2y^2 + y) dA = \int_{-\pi/2}^{\pi/2} \int_0^{\cos(\theta)} \underline{\hspace{2cm}} dr d\theta$$

6. Determine the volume of the region that lies under the sphere $x^2 + y^2 + z^2 = 9$, above the plane $z = 0$ and inside the cylinder $x^2 + y^2 = 5$.

7. Use a double integral to determine the area of the region bounded by the curves $y = \sqrt{x}$ and $y = x^2 + 1$ on the interval from $[1, 2]$.

8. Evaluate the integral $\int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \cos(x^2 + y^2) dy dx$.